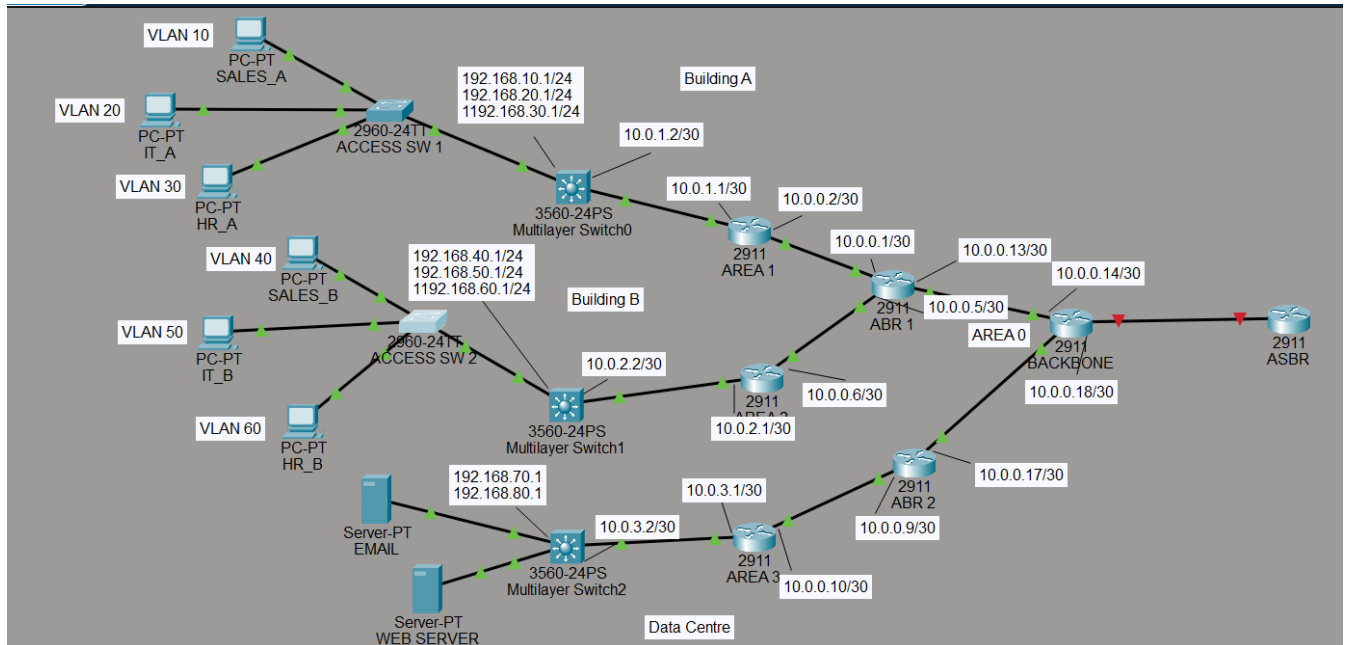


ENTERPRISE PROJECT 3

VLAN + MULTI- AREA OSPF + SECURITY (ACL, DHCP + DHCP SNOOPING, DAI & PORT SECURITY) + DUAL STACK

DIAGRAM:



DESCRIPTION:

This project involves designing a secure and scalable enterprise network. VLANs are used to divide the network into separate departments, while multi-Area OSPF enables efficient routing between different parts of the network. Security features such as ACLs, DHCP with DHCP Snooping, Dynamic ARP Inspection (DAI), and Port Security are implemented to prevent unauthorized access and network attacks. The network also runs in Dual Stack mode, supporting both IPv4 and IPv6 for future readiness.

Note: This project is just simulation of enterprise network, it might vary from real world setup.

STEP 1: Create VLANs, Access Ports & Trunk:

Create Vlan 10, Vlan 20, Vlan 30, Vlan 40, Vlan 50, Vlan 60, Vlan 70 & Vlan 80 on all switches.
Enable access port for each PC port based on VLAN for relevant Switches.

allow trunks on required paths.

STEP 2: Configure L3 Switch (AREA 1):

ip routing

interface vlan 10

ip address 192.168.10.1 255.255.255.0

no shutdown

interface vlan 20

ip address 192.168.20.1 255.255.255.0

no shutdown

interface vlan 30

ip address 192.168.30.1 255.255.255.0

no shutdown

configure uplink to AREA 1 router:

interface g0/1

no switchport

ip address 10.0.1.2 255.255.255.252

no shutdown

ip dhcp excluded-address 192.168.10.1 192.168.10.10

ip dhcp excluded-address 192.168.20.1 192.168.20.10

ip dhcp excluded-address 192.168.30.1 192.168.30.10

ip dhcp pool SALES_A

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

dns-server 8.8.8.8

ip dhcp pool IT_A

network 192.168.20.0 255.255.255.0

default-router 192.168.20.1

dns-server 8.8.8.8

ip dhcp pool HR_A

network 192.168.30.0 255.255.255.0

default-router 192.168.30.1

dns-server 8.8.8.8 (optional)

```
router ospf 1
router-id 1.1.1.10
network 192.168.10.0 0.0.0.255 area 1
network 192.168.20.0 0.0.0.255 area 1
network 192.168.30.0 0.0.0.255 area 1
network 10.0.1.0 0.0.0.3 area 1
```

```
passive-interface vlan 10(optional)
passive-interface vlan 20
passive-interface vlan 30
```

Note: If trunk fails in L3, use encapsulation dot1q command to disable auto

STEP 3 : Configure AREA 1 router (2911):

Link to L3 switch:

```
interface g0/1
ip address 10.0.1.1 255.255.255.252
no shutdown
```

Link to ABR1:

```
interface g0/2
ip address 10.0.0.2 255.255.255.252
no shutdown
```

```
router ospf 1
router-id 1.1.1.1
network 10.0.1.0 0.0.0.3 area 1
network 10.0.0.0 0.0.0.3 area 0
```

STEP 4: Configure L3 Switch(AREA 2):

Ip routing

```
interface vlan 40
ip address 192.168.40.1 255.255.255.0
no shutdown
```

```
interface vlan 50
```

```
ip address 192.168.50.1 255.255.255.0
no shutdown
```

```
interface vlan 60
ip address 192.168.60.1 255.255.255.0
no shutdown
```

uplink to Aea 2 router:

```
interface g0/1
no switchport
ip address 10.0.2.2 255.255.255.252
no shutdown
```

```
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.50.1 192.168.50.10
ip dhcp excluded-address 192.168.60.1 192.168.60.10
```

```
ip dhcp pool SALES_B
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 8.8.8.8
```

```
ip dhcp pool IT_B
network 192.168.50.0 255.255.255.0
default-router 192.168.50.1
dns-server 8.8.8.8
```

```
ip dhcp pool HR_B
network 192.168.60.0 255.255.255.0
default-router 192.168.60.1
dns-server 8.8.8.8
```

```
router ospf 1
router-id 2.2.2.20
network 192.168.40.0 0.0.0.255 area 2
network 192.168.50.0 0.0.0.255 area 2
network 192.168.60.0 0.0.0.255 area 2
network 10.0.2.0 0.0.0.3 area 2
```

```
passive-interface vlan 40
passive-interface vlan 50
passive-interface vlan 60
```

STEP 5: Configure Area 2 router (2911):

Link to L3 switch:

```
interface g0/1
ip address 10.0.2.1 255.255.255.252
no shutdown
```

Link to ABR1:

```
interface g0/0
ip address 10.0.0.6 255.255.255.252
no shutdown
```

```
router ospf 1
router-id 2.2.2.2
network 10.0.2.0 0.0.0.3 area 2
network 10.0.0.4 0.0.0.3 area 0
```

STEP 6 : Configure L3 switch(Area 3):

```
interface vlan 70
ip address 192.168.70.1 255.255.255.0
no shutdown
```

```
interface vlan 80
ip address 192.168.80.1 255.255.255.0
no shutdown
```

Uplink to area 3 router:

```
interface g0/1
no switchport
ip address 10.0.3.2 255.255.255.252
no shutdown
```

```
router ospf 1
router-id 3.3.3.30
network 192.168.70.0 0.0.0.255 area 3
network 192.168.80.0 0.0.0.255 area 3
network 10.0.3.0 0.0.0.3 area 3
```

```
passive-interface vlan 70 (optional)
passive-interface vlan 80
```

STEP 7 : Configure area 3 router (2911):

Link to L3 switch:

```
interface g0/1
ip address 10.0.3.1 255.255.255.252
no shutdown
```

link to abr2 router:

```
interface g0/0
ip address 10.0.0.10 255.255.255.252
no shutdown
```

```
router ospf 1
router-id 3.3.3.3
network 10.0.3.0 0.0.0.3 area 3
network 10.0.0.8 0.0.0.3 area 0
```

STEP 8: Configure ABR1 router:

```
interface g0/2
ip address 10.0.0.1 255.255.255.252
no shutdown
```

```
interface g0/0
ip address 10.0.0.5 255.255.255.252
no shutdown
```

```
interface g0/1
ip address 10.0.0.13 255.255.255.252
no shutdown
```

```
router ospf 1
router-id 11.11.11.11
```

```
network 10.0.0.0 0.0.0.3 area 0
network 10.0.0.4 0.0.0.3 area 0
network 10.0.0.12 0.0.0.3 area 0
```

STEP 9: Configure ABR2 router:

```
interface g0/0
ip address 10.0.0.9 255.255.255.252
no shutdown
```

```
interface g0/1
ip address 10.0.0.17 255.255.255.252
no shutdown
```

```
router ospf 1
router-id 22.22.22.22
network 10.0.0.8 0.0.0.3 area 0
network 10.0.0.16 0.0.0.3 area 0
```

STEP 10: Configure BACKBONE router:

```
interface g0/1
ip address 10.0.0.14 255.255.255.252
no shutdown
```

```
interface g0/2
ip address 10.0.0.18 255.255.255.252
no shutdown
```

```
router ospf 1
router-id 0.0.0.1
```

```
network 10.0.0.12 0.0.0.3 area 0
network 10.0.0.16 0.0.0.3 area 0
```

STEP 11: Configure DHCP snooping & DAI:

On access switch(area 1):

```
ip dhcp snooping
ip dhcp snooping vlan 10,20,30
ip arp inspection vlan 10,20,30
```

```
int fa0/24
ip dhcp snooping trust
ip arp inspection trust
```

on access switch(area 2):

```
ip dhcp snooping
ip dhcp snooping vlan 40,50,60
ip arp inspection vlan 40,50,60
```

```
int fa0/24
ip dhcp snooping trust
ip arp inspection trust
```

STEP 12: Configure Standard & Extended:

Block HR pcs from accessing Web server(Standard Number ACL):

On L3 switch(area 3, close to destination):

```
access-list 10 deny 192.168.30.0 0.0.0.255
access-list 10 deny 192.168.60.0 0.0.0.255
access-list 10 permit any
```

```
interface vlan 80
ip access-group 10 in
```

Block ICMP for sales PC(Named Extended ACL):

On L3 switch (Area 1, close to source):

```
ip access-list extended SALES_BLOCK_ICMP
deny icmp 192.168.10.0 0.0.0.255 host 192.168.80.10 echo
deny icmp 192.168.10.0 0.0.0.255 host 192.168.70.10 echo
permit ip any any
```

```
interface vlan 10
ip access-group SALES_BLOCK_ICMP in
```

Area 2 L3 switch:

```
ip access-list extended SALES_BLOCK_ICMP
deny icmp 192.168.40.0 0.0.0.255 host 192.168.80.10 echo
deny icmp 192.168.40.0 0.0.0.255 host 192.168.70.10 echo
permit ip any any
```

```
interface vlan 40
ip access-group SALES_BLOCK_ICMP in
```


STEP 13: PCs and server configuration:

PC	VLAN	IP Address	Gateway/CIDR
SALES_A	10	192.168.10.11	192.168.10.1
IT_A	20	192.168.20.11	192.168.20.1
HR_A	30	192.168.30.11	192.168.30.1
SALES_B	40	192.168.40.11	192.168.40.1
IT_B	50	192.168.50.11	192.168.50.1
HR_B	60	192.168.60.11	192.168.60.1
EMAIL SERVER	70	192.168.80.10	192.168.70.1
WEB SERVER	80	192.168.70.10	192.168.80.1
CIDR/SUBNETMASK	/24	255.255.255.0	

STEP 14: Configure ipv6(optional):

L3 Switch (Area 1):

```
interface vlan 10
ipv6 address 2001:DB8:10::1/64
ipv6 ospf 1 area 1
```

```
interface vlan 20
ipv6 address 2001:DB8:20::1/64
ipv6 ospf 1 area 1
```

```
interface vlan 30
ipv6 address 2001:DB8:30::1/64
ipv6 ospf 1 area 1
```

```
interface g0/1
ipv6 address 2001:DB8:1::2/64
ipv6 ospf 1 area 1
```

```
ipv6 router ospf 1
router-id 1.1.1.10
```

area 1 router:

```
interface g0/1
  ipv6 address 2001:DB8:1::1/64
  ipv6 ospf 1 area 1

interface g0/2
  ipv6 address 2001:DB8:0:1::2/64
  ipv6 ospf 1 area 0

ipv6 router ospf 1
  router-id 1.1.1.1
```

L3 switch (area 2):

```
interface vlan 40
  ipv6 address 2001:DB8:40::1/64
  ipv6 ospf 1 area 2

interface vlan 50
  ipv6 address 2001:DB8:50::1/64
  ipv6 ospf 1 area 2

interface vlan 60
  ipv6 address 2001:DB8:60::1/64
  ipv6 ospf 1 area 2

interface g0/1
  ipv6 address 2001:DB8:2::2/64
  ipv6 ospf 1 area 2

ipv6 router ospf 1
  router-id 2.2.2.20
```

area 2 router:

```
interface g0/1
  ipv6 address 2001:DB8:2::1/64
  ipv6 ospf 1 area 2

interface g0/0
  ipv6 address 2001:DB8:0:2::2/64
  ipv6 ospf 1 area 0
```

```
ipv6 router ospf 1
router-id 2.2.2.2
```

L3 switch (area 3):

```
interface vlan 70
ipv6 address 2001:DB8:70::1/64
ipv6 ospf 1 area 3
```

```
interface vlan 80
ipv6 address 2001:DB8:80::1/64
ipv6 ospf 1 area 3
```

```
interface g0/1
ipv6 address 2001:DB8:3::2/64
ipv6 ospf 1 area 3
```

```
ipv6 router ospf 1
router-id 3.3.3.30
```

area 3 router:

```
interface g0/1
ipv6 address 2001:DB8:3::1/64
ipv6 ospf 1 area 3
```

```
interface g0/0
ipv6 address 2001:DB8:0:3::2/64
ipv6 ospf 1 area 0
```

```
ipv6 router ospf 1
router-id 3.3.3.3
```

ABR1 router:

```
interface g0/2
ipv6 address 2001:DB8:0:1::1/64
ipv6 ospf 1 area 0
no shutdown
```

```
interface g0/2
ipv6 address 2001:DB8:0:2::1/64
ipv6 ospf 1 area 0
no shutdown
```

```
interface g0/1
```

```
ipv6 address 2001:DB8:0:4::1/64
ipv6 ospf 1 area 0
no shutdown
```

```
ipv6 router ospf 1
router-id 11.11.11.11
```

ABR2 router:

```
interface g0/0
ipv6 address 2001:DB8:0:3::1/64
ipv6 ospf 1 area 0
no shutdown
```

```
interface g0/1
ipv6 address 2001:DB8:0:4::2/64
ipv6 ospf 1 area 0
no shutdown
```

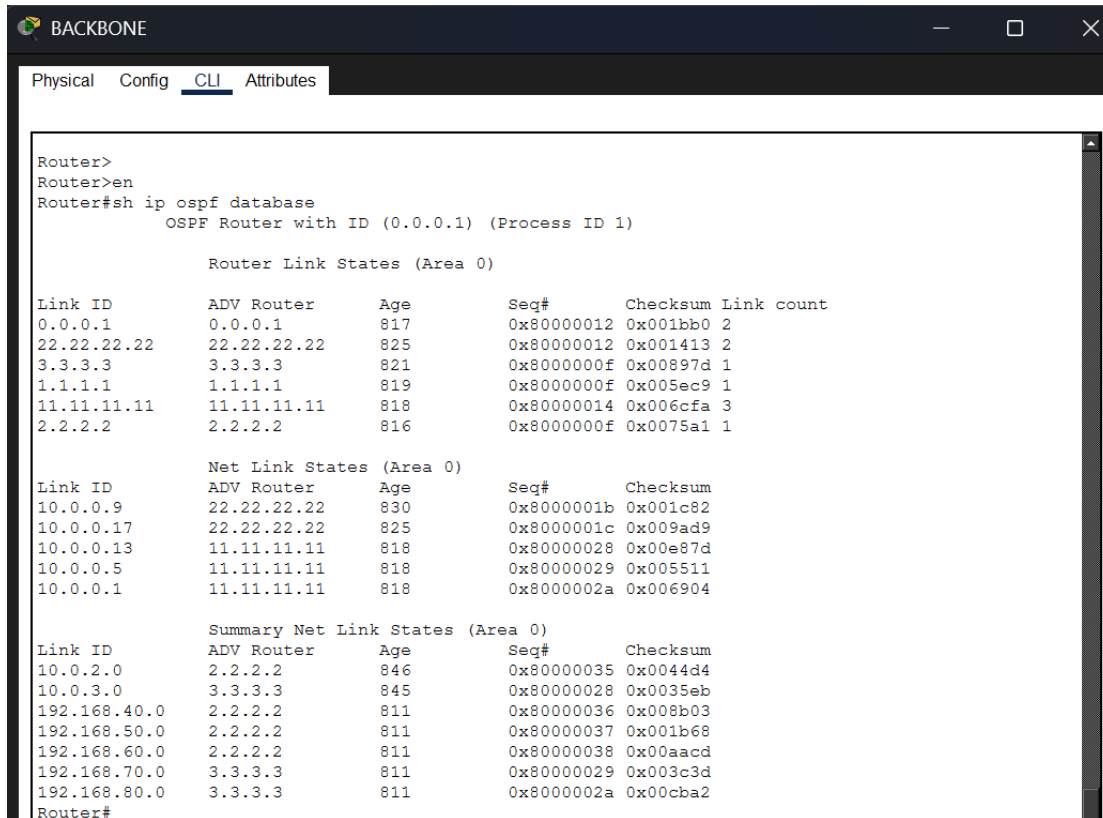
```
ipv6 router ospf 1
router-id 22.22.22.22
```

STEP 15: Verifications & Ping Test:

```
show ip ospf neighbor
show ip ospf database
show ip route ospf
show ip ospf interface brief
show vlan brief
show interfaces trunk
show interfaces switchport
show ip route
show ip interface brief
show ip dhcp pool
show ip dhcp binding
show ip dhcp server statistics
show ip dhcp snooping
show ip dhcp snooping binding
show ip arp inspection
show ip arp inspection statistics
show access-lists
show run | include access-group
show ipv6 interface brief
show ipv6 route
show ipv6 ospf neighbor
```

OSPF BACKBONE DATABASE:

Backbone router contains all summarized routes of all the area(not full details of type 1 & type 2 LSAs of other areas),



```
Router>
Router>en
Router#sh ip ospf database
      OSPF Router with ID (0.0.0.1) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age      Seq#          Checksum Link count
0.0.0.1        0.0.0.1       817      0x80000012   0x001bb0 2
22.22.22.22    22.22.22.22   825      0x80000012   0x001413 2
3.3.3.3        3.3.3.3       821      0x8000000f   0x00897d 1
1.1.1.1        1.1.1.1       819      0x8000000f   0x005ec9 1
11.11.11.11    11.11.11.11   818      0x80000014   0x006cfa 3
2.2.2.2        2.2.2.2       816      0x8000000f   0x0075a1 1

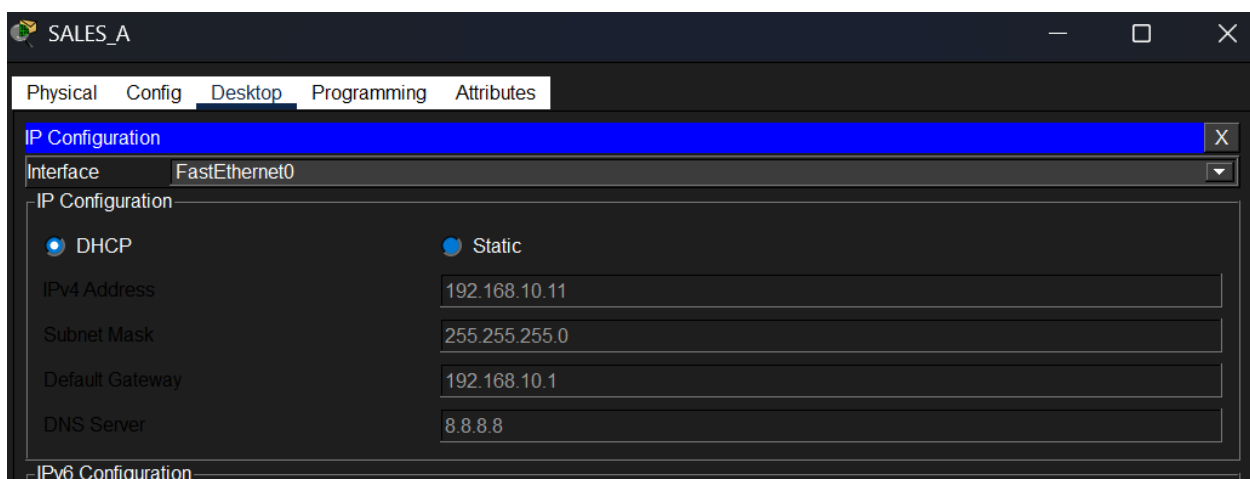
      Net Link States (Area 0)

Link ID        ADV Router    Age      Seq#          Checksum
10.0.0.9       22.22.22.22   830      0x8000001b   0x001c82
10.0.0.17      22.22.22.22   825      0x8000001c   0x009ad9
10.0.0.13      11.11.11.11   818      0x80000028   0x00e87d
10.0.0.5       11.11.11.11   818      0x80000029   0x005511
10.0.0.1       11.11.11.11   818      0x8000002a   0x006904

      Summary Net Link States (Area 0)

Link ID        ADV Router    Age      Seq#          Checksum
10.0.2.0       2.2.2.2       846      0x80000035   0x0044d4
10.0.3.0       3.3.3.3       845      0x80000028   0x0035eb
192.168.40.0    2.2.2.2       811      0x80000036   0x008b03
192.168.50.0    2.2.2.2       811      0x80000037   0x001b68
192.168.60.0    2.2.2.2       811      0x80000038   0x00aacd
192.168.70.0    3.3.3.3       811      0x80000029   0x003c3d
192.168.80.0    3.3.3.3       811      0x8000002a   0x00cba2
Router#
```

DHCP is enabled:



Ping test:

Ping from SALES PC to SERVER(Extended ACL blocked):

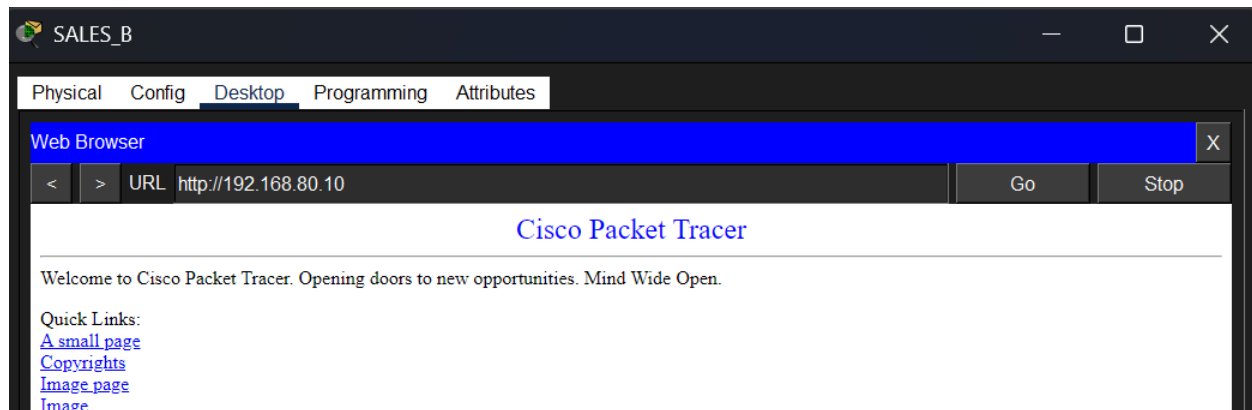
```
C:\>ping 192.168.70.10

Pinging 192.168.70.10 with 32 bytes of data:

Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.

Ping statistics for 192.168.70.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Browse from SALES PC:



Ping test from IT_B PC to SALES_B PC:

```
C:\>ping 192.168.40.11

Pinging 192.168.40.11 with 32 bytes of data:

Reply from 192.168.40.11: bytes=32 time=16ms TTL=127
Reply from 192.168.40.11: bytes=32 time=11ms TTL=127
Reply from 192.168.40.11: bytes=32 time=10ms TTL=127
Reply from 192.168.40.11: bytes=32 time<1ms TTL=127
```

NOTE: ipv6 is not implement due to some limitations in PT. ipv6 used OSPFv3 and can be implemented in real devices. This project simulation of real world multi-area OSPF and not follows exact topology.